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Cerebrovascular segmentation in phase-contrast magnetic resonance angiography by a Radon projection composition network

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ABSTRACT

Cerebrovascular segmentation based on phase-contrast magnetic resonance angiography (PC-MRA) provides patient-specific intracranial vascular structures for neurosurgery planning. However, the vascular complex topology and spatial sparsity make the task challenging. Inspired by the computed tomography reconstruction, this paper proposes a Radon Projection Composition Network (RPC-Net) for cerebrovascular segmentation in PC-MRA, aiming to enhance distribution probability of vessels and fully obtain the vascular topological information. Multi-directional Radon projections of the images are introduced and a two-stream network is used to learn the features of the 3D images and projections. The projection domain features are remapped to the 3D image domain by filtered back-projection transform to obtain the image-projection joint features for predicting vessel voxels. A four-fold cross-validation experiment was performed on a local dataset containing 128 PC-MRA scans. The average Dice similarity coefficient, precision and recall of the RPC-Net achieved 86.12%, 85.91% and 86.50%, respectively, while the average completeness and validity of the vessel structure were 85.50% and 92.38%, respectively. The proposed method outperformed the existing methods, especially with significant improvement on the extraction of small and low-intensity vessels. Moreover, the applicability of the segmentation for electrode trajectory planning was also validated. The results demonstrate that the RPC-Net realizes an accurate and complete cerebrovascular segmentation and has potential applications in assisting neurosurgery preoperative planning.

1. Introduction

Intracranial hemorrhage is one of the main risks of minimally invasive stereotactic neurosurgery such as deep brain stimulation, stereo-electroencephalography and laser thermal ablation. Constructing an accurate vascular network based on intracranial angiography for further preoperative planning is significant for avoiding the risk (Scorza et al., 2021). Magnetic resonance angiography (MRA) has become one of the most used clinical vascular images due to its high resolution and no radiation advantages (Xiao et al., 2017). Among the different MRAs, phase-contrast MRA (PC-MRA) can simultaneously image arteries and veins and suppress non-vascular tissues without contrast agents, thus being suitable for cerebrovascular structure modeling (Chen et al., 2022; Du et al., 2013). However, PC-MRA is susceptible to noise and the signal of vessels is nonuniform. Intracranial vessels have sparse spatial distribution and complex topology. These all make it challenging to extract intracranial vessels in PC-MRA automatically.

Automatic cerebrovascular segmentation based on medical images has always been a hotspot. The related methods mainly focus on the brain arteries in time-of-flight MRA (TOF-MRA), which can be divided into traditional and deep learning methods. Traditional segmentation methods generally utilize artificially selected features (Frangi et al., 1998; Jerman et al., 2016; Law and Chung, 2008) and formulate corresponding rules to distinguish vessels from the background, such as threshold methods (Dufour et al., 2013; Wang et al., 2015), parametric model methods (Forkert et al., 2013; Zhao et al., 2014), and statistical model methods (Lu et al., 2016; Zhang et al., 2019; Zhou et al., 2019). However, traditional methods rely on artificially formulated features, rules and hyperparameters, making it difficult to achieve good accuracy and robustness for PC-MRA images with large variations. In contrast, deep learning methods represented by convolutional neural networks (CNN) can automatically extract features of vessels and have been applied to cerebrovascular segmentation with excellent performance in recent years. The U-Net (Çiçek et al., 2016; Ronneberger et al., 2015)

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and its variants are the widely used network architectures. Livne et al. used 2D U-Net to segment intracranial arteries in TOF-MRA slices (Livne et al., 2019). Guo et al. used three 2D U-Nets to segment vessels in three cross-sectional images of TOF-MRA (Guo et al., 2021). To fully utilize the 3D information of vessels in 3D images, Tetteh et al. proposed DeepVesselNet, which uses a 2D crosshair filter to approximate the operation of the 3D convolution kernel, reducing the calculation complexity and memory spend (Tetteh et al., 2020). Sanchesa et al. proposed the Uception, which introduced the Inception module in 3D U-Net to extract multi-scale features (Sanchesa et al., 2019). Hilbert et al. proposed BRAVE-NET, which adopted a multi-scale supervised learning strategy based on 3D U-Net (Hilbert et al., 2020). The above methods have achieved good results, but there are still deficiencies for the brain vessels with complex topology and sparse distribution. In this regard, several studies have been devoted to enhancing the network's ability to extract vascular topological information. Xia et al. introduced a reverse edge attention module in 3D U-Net to extract edge features that can characterize topology (Xia et al., 2022). Wang et al. proposed JointVesselNet and VC-Net, which introduced the maximum intensity projection (MIP) (Wang et al., 2020; Wang et al., 2021). The networks extracted features from 3D image volumes and 2D MIP, and then performed feature fusion through back-projection transform. The methods applied the principle of image volume rendering to enhance the vascular topological representation and used MIP to enhance the distribution probability of vascular voxels in the network input.

The above studies have demonstrated the advantages of deep learning in the cerebrovascular segmentation task, but its application in PC-MRA still needs to be verified and further improved. Only a few research currently have focused on cerebrovascular segmentation in PC-MRA. Some studies (Du et al., 2013; Xiao et al., 2017; Zuluaga et al., 2015) enhanced the shape characteristics with vesselness filters and distinguished the vessels and background by intensity distribution. However, the segmentation accuracy is undesirable. Chen et al. first applied CNN to cerebrovascular segmentation in PC-MRA, using multiple networks to extract deep features and fusing the segmentation results based on the Dempster-Shafer evidence theory (Chen et al., 2022). However, this study was only conducted on a small dataset, so its generalization needs further verification. Moreover, the performance still needs to be improved due to the lack of consideration of complex topology and distribution sparsity of brain vessels.

Inspired by the computed tomography (CT) reconstruction algorithm, this paper proposes a Radon Projection Composition Network (RPC-Net) for cerebrovascular segmentation in PC-MRA. The method uses a two-stream CNN framework to extract features from the 3D images and multi-directional Radon projections. The features in the projection domain are remapped to the 3D image domain through back-projection Radon transform and then fused with the features in the original image domain to obtain joint features and predict cerebrovascular voxels. The further experiments demonstrate that the proposed method realizes an accurate and complete segmentation of intracranial vessels in PC-MRA, and obtains better segmentation results than the existing methods on vessels of different sizes and intensities.

2. Method

2.1. Overview

This paper develops a new cerebrovascular segmentation method, which combines 3D volume image information with multi-directional projection information to mine complex topological features of intracranial vessels. The 3D image can accurately reflect the 3D spatial structure of vessels, but the image noise, low contrast and distribution sparsity of vessels interfere with vessel segmentation. Single-directional MIP used in VC-Net enhances the topology and connectivity of vessels. However, spatial information is lost and the coverage problem between vessels may occur. This work integrates the advantages of 3D image and multi-directional projection, and realizes the mutual conversion between 3D image domain and projection domain through the Radon transform and its inverse transform. As shown in Fig. 1, the vessels in the images are sparsely distributed in the 3D space, and their connectivity in different directions exhibits large variability due to the complex geometries. In contrast, the Radon projection of each direction fuses multiple slices to characterize the vasculature shape, enhancing the vascular continuity and geometric representation and adapting to the vascular geometric variability. The proposed method adopts a patch-wise segmentation strategy with three steps: image preprocessing, vessel segmentation using RPC-Net and patch label fusion, which will be introduced below.

2.2. Image preprocessing

In the image preprocessing step, the intensity of the PC-MRA image is mapped to [0,1] by the maximum-minimum normalization method. In order to avoid the impact of resolution anisotropy on projection generation, the minimum resolution of the three orthogonal axes is used as the uniform resolution to resample each 3D image. As RPC-Net takes image patches as input, the original image is randomly cropped into patches with the size of $64 \times 64 \times 64$ for network training. During the testing and inference phase, the original image is cropped into $64 \times 64 \times 64$ patches with a stride of 16 voxels.

2.3. RPC-Net

2.3.1. Network architecture

This work uses a two-stream network (Fig. 2) like VC-Net. The network contains a 3D image segmentation stream and a projection segmentation stream, which take the image patches and the corresponding multi-directional projections as input, respectively. First, two 3D U-Nets are used to extract the features through the encoder and decoder. The features output by the two last layers are regarded as the features of the 3D image domain and the projection domain, respectively. Then, the back-projection Radon transform is applied to map the features of the projection domain back to the image domain. Next, the features of both domains are stacked to form the joint features, which are finally used to predict the vessel voxels. Meanwhile, the projection segmentation stream outputs the vessel segmentation in the projection image and is supervised for optimization. Different from VC-Net, this paper uses 3D U-Net instead of 2D U-Net in the projection segmentation stream, aiming to exploit the intrinsic correlation between projection images of adjacent angles in dense projection directions.

2.3.2. Multi-directional Radon projection

Radon transform is performed on the original image patches to get multi-directional projections. The proposed method selects 96 projection directions parallel to the cross-section, which rotate along the vertical axis of each patch and divide the semicircle evenly into 96 parts. The multi-directional projections avoid the problem of high-intensity vessels covering low-intensity vessels, which may occur in single-direction projection. In addition, the intensity integral is equivalent to performing mean filtering on the image, which helps suppress random noise.

The Radon transform is the basic transform applied in CT reconstruction, which integrates the intensity of the image along multiple lines (Fig. 3(A)). The transform used in this paper is to perform parallel beam projection on each cross-sectional image. 1D projection data is obtained after transformation, and the projection data of all cross-sectional images and angles in the semicircle constitutes the final projection. Given the 3D image volume I, the Radon projection of the cross-section $z \in \{0, 1, \dots, Z-1\}$along the direction $\theta \in [0, 180)$ is calculated as:

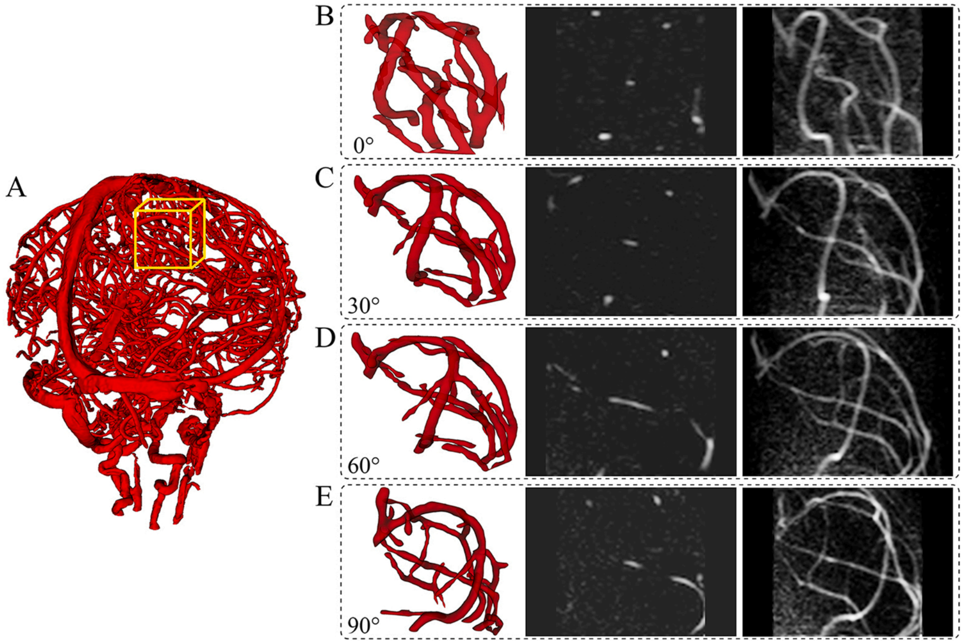

Fig. 1. The illustration of the multi-directional Radon projections. A: The 3D cerebrovascular model with a yellow box marking the image patch to be projected. B-E: The rotated vascular models (left), rotated central slices (middle) and Radon projections (right) of the image patch at orientations of 0, 30, 60 and 90 degrees, respectively.

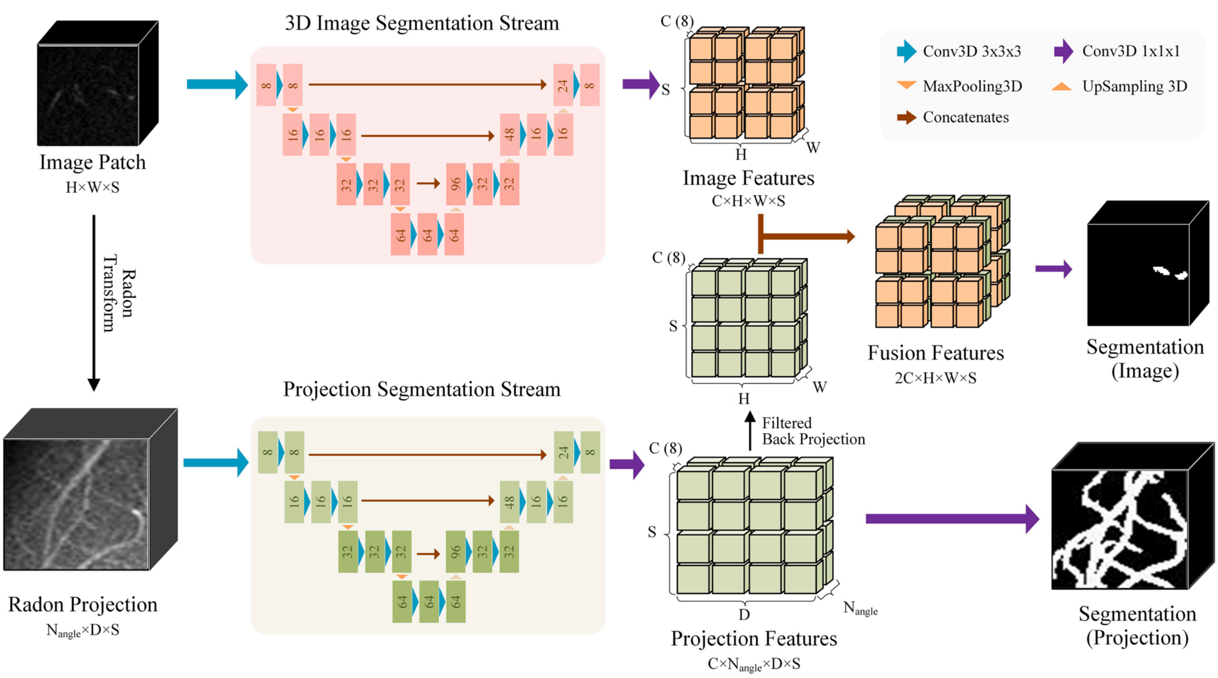

Fig. 2. The architecture of RPC-Net.

$$\mathscr{R}_I(z,\theta,R)=\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}I(x,y,z)\bullet\delta(x\cos\theta+y\sin\theta-R)dxdy, \tag{1}$$

where $x\cos\theta+y\sin\theta-R$ represents the projection line, $\delta(\bullet)$ is the impulse function, and R is defined as the axis coordinate of the one-dimensional projection data. In the discrete circumstance, the image is represented as a square grid with pixel cells assigned the intensity constants, and each projection ray becomes a strip with the width of projection spacing. The line integral of each projection ray is calculated as the sum of intensity weighted by each fractional area of the pixel cell intercepted by the ray strip. Since there are multiple down-sampling operations in the network, the maximum value of R is $\sqrt{2}$ times the image patch width W, the value of R is set to $\{0,1,\dots,R_{\max}\}$ and $R_{\max}$ is defined as

$$R_{\max}=\left(\left[\frac{W*\sqrt{2}}{8}\right]+1\right)*8, \tag{2}$$

where $[*]$ represents rounding operation.

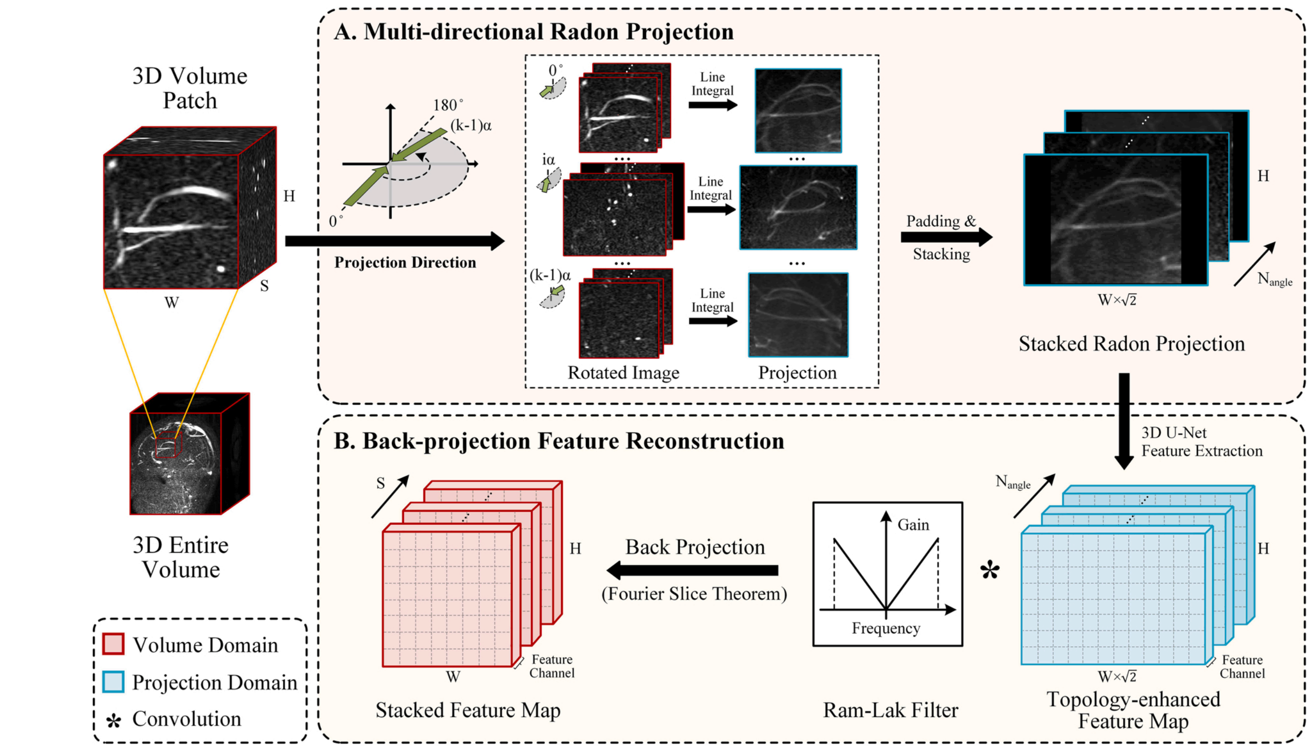

Fig. 3. Radon projection and back-projection reconstruction. A. Illustration of the image-to-projection process via radon transform. B. Illustration of the projection-to-image process via filtered back projection.

2.3.3. *Back-projection reconstruction of Features*

The dimensions of the projection image represent the projection axis, projection direction and cross-section index, which are different from the original image. Therefore, the features of the projection domain need to be back-projected to the 3D image domain. Direct mapping from the projection domain to the image domain is difficult to establish. Thus, this work uses the standard process of CT reconstruction based on the Fourier slice theorem and constructs the indirect transfer relation of Projection-Fourier-Image space. The Fourier slice theorem indicates that the 1D Fourier transform of the Radon projection is equal to a straight line passing through the origin with angle θ in the 2D Fourier space of the image:

$$\mathscr{F}(\rho,\theta,z) = \mathscr{F}_1\{R_I(z,\theta,R)\}, \tag{3}$$

where $\mathscr{F}_1$ represents the 1D Fourier transform and ρ is the radial coordinate of the polar coordinate system of the image domain. Accordingly, in order to realize the feature back projection, 1D Fourier transform is performed on the projection data of each direction to fill the Fourier space, and then an inverse 2D Fourier transform is performed to finish the conversion from the projection domain into the image domain. To avoid the artifacts caused by the above direct back-projection process, we use the convolutional back-projection algorithm (Fig. 3(B)) as follows:

$$f(x,y,z) = \int_0^{2\pi} d\theta \int_{-\infty}^{+\infty} [R_I(z,\theta,R) * C(R)]\delta(x\cos\theta + y\sin\theta - R)dR, \tag{4}$$

where $C(R)$ represents the Ram-Lak filter, which can help retain the high-frequency components in the feature map:

$$C(R) = \frac{1}{4\Delta p^2}\left[2\mathrm{sinc}\left(\frac{R}{\Delta p}\right) - \mathrm{sinc}^2\left(\frac{R}{2\Delta p}\right)\right], \tag{5}$$

where $\mathrm{sinc}(x)$ stands for $\sin(\pi x)/(\pi x)$ and Δp is the sampling interval.

2.3.4. *Loss function*

The loss function contains two parts, namely the 3D image segmentation loss L_{img} and the projection segmentation loss L_{proj}:

$$L_{\mathrm{total}} = \alpha L_{\mathrm{img}} + \beta L_{\mathrm{proj}}, \tag{6}$$

where α and β are constant coefficients, set to be 0.8 and 0.2 for our best experiment performances, respectively. Both L_{img} and L_{proj} use Dice loss that is computed as

$$\text{Dice Loss} = 1 - \frac{2\sum_{x\in V} p(x)g(x) + \delta}{\sum_{x\in V} p(x) + \sum_{x\in V} g(x) + \delta}, \tag{7}$$

where V represents the segmented image, $p(x)$ and $g(x)$ denote the prediction result and the ground truth for voxel x in V, respectively. δ is a small smooth constant. For the Dice loss used as L_{img} and L_{proj}, V represents the input image patch and the corresponding projection, respectively.

2.4. *Patch Label fusion*

Since vessel truncation may occur at the edge of the patches and the projected thickness of the patch border is small in some directions, the segmentation accuracy at the boundary is low. Therefore, for each $64 \times 64 \times 64$ patch, the central $48 \times 48 \times 48$ voxels are given a weight of 1, and the other edge voxels are given a weight of 0.5. Then the weighted probability of each voxel belonging to the vessel class is calculated to determine the final category. Given a voxel v, its final class is calculated from the segmentation results of the k patches $(S_1, S_2, \ldots, S_k)$ containing v as

$$S(v) = \left(\frac{\sum_{i=1}^{k} w_i(v) S_i(v)}{\sum_{i=1}^{k} w_i(v)}\right) > 0.5, \tag{8}$$

where $S_i(v)$ denotes the probability that voxel v is vessel in the i-th patch and $w_i(v)$ is the weight.

3. Experiments and results

3.1. Dataset

This study included 128 clinically collected PC-MRA scans from Tsinghua University Imaging Center, Peking University First Hospital, The Second Hospital of Hebei Medical University and Beijing RIMAG Imaging Center. The data was cleaned to remove patient privacy. The dataset consisted of 71 males and 57 females, ranging in age from 3 to 62 years (average of 21.6 years). Image acquisition parameters are shown in Table 1. The image slice size ranged from 288 × 288–512 × 512, with the slice number ranging from 170 to 367. The intra-slice resolution ranged from 0.449 mm to 0.799 mm, and the slice spacing ranged from 0.6 mm to 0.8 mm. The ground truth of intracranial vessels was identified and annotated by experienced clinicians. The scalp vessels in the images were not included in the manual annotations as they are not the vessels of interest for stereotactic neurosurgery.

3.2. Implementation detail

Four-fold cross validation was used for dataset partitioning and method validation. The networks were implemented in PyTorch framework and trained on a platform with two Intel(R) Xeon(R) Gold 6248R CPUs and four NVIDIA RTX3090 24 GB GPUs. Each network was trained for 80 epochs using SGD optimizer with a batch size of 64 and a momentum of 0.9. The initial learning rate was set to 0.1 and the L2 regularization factor was set to 1e-8. The Radon transform and filtered back-projection used in the method were realized by the open-source library TorchRadon (Ronchetti, 2020), and the parameter det_spacing that represents the distance between two contiguous rays was set to 1.0 pixel. The ground truth of the projection segmentation was generated by applying the Radon transform to the ground truth of image segmentation.

3.3. Evaluation metric

This work adopted the Dice similarity coefficient (DSC), Precision and Recall as indexes to assess the segmentation performance:

$$\mathrm{DSC} = \frac{2|P \cap G|}{|P| + |G|}, \tag{9}$$

$$\mathrm{Precision} = \frac{|P \cap G|}{|P|}, \tag{10}$$

Table 1
Image acquisition parameters of the dataset.

Center	Scan Num.	Manufacturer	Repetition Time (ms)	Flip Angle	Echo Time (ms)
Tsinghua University Imaging Center	91	Philips Medical Systems (Achieva)	20	15	4.325–10.139
Peking University First Hospital	13	Philips Medical Systems (Achieva)	20	15	7.743–7.895
The Second Hospital of Hebei Medical University	20	Philips Medical Systems (Achieva)	10	10	6.926–7.274
Beijing RIMAG Imaging Center	4	GE Medical Systems (SIGNA Pioneer)	16	20	5.288–5.332

$$\mathrm{Recall} = \frac{|P \cap G|}{|G|}, \tag{11}$$

where P and G denote the prediction and the ground truth, respectively. $|*|$ represents the number of voxels and $\cap$ is the intersection operation.

The accuracy of the extracted vessel network significantly influences vascular structure analysis and surgical planning. Therefore, this paper additionally used the completeness and validity index for the skeleton (Tang et al., 2021) to evaluate the segmentation performance. First, the hole-filling morphological operation was performed on the prediction and the ground truth, and then the vessel skeleton was extracted by the vessel thinning algorithm (Lee et al., 1994). To prevent evaluation errors caused by skeleton extraction bias, the vascular skeleton extracted from the ground truth was dilated using a structural element with a radius of 2 mm. Finally, the completeness and validity of the skeleton were calculated as follows:

$$\mathrm{Completeness} = \frac{|S_{\mathrm{pred}} \cap D(S_{\mathrm{GT}})|}{|S_{\mathrm{GT}}|}, \tag{12}$$

$$\mathrm{Validity} = \frac{|S_{\mathrm{pred}} \cap D(S_{\mathrm{GT}})|}{|S_{\mathrm{pred}}|}, \tag{13}$$

where S_{pred} and S_{GT} denote the vascular skeleton extracted from the prediction and the ground truth, respectively, and $D(*)$ represents the dilation operation. As the completeness and validity indexes are similar to the Recall and Precision, we calculated the F1 Score for the skeleton extraction based on the formula of the F1 Score for classification tasks:

$$\mathrm{F1Score} = \frac{2 * \mathrm{Completeness} * \mathrm{Validity}}{\mathrm{Completeness} + \mathrm{Validity}} \tag{14}$$

The F1 Score combines completeness and validity to evaluate the accuracy of the vascular structure comprehensively.

3.4. Comparison with other methods

3D U-Net, BRAVE-Net and VC-Net were selected as reference methods. 3D U-Net is the baseline model, BRAVE-Net is the representative of the U-Net variant model, and VC-Net is the method that also used projection ideas. All reference methods were re-implemented and tested on the local PC-MRA dataset. The comparison between methods on the accuracy of cerebrovascular segmentation is shown in Table 2. The results show that RPC-Net was superior to other methods, obtaining the highest DSC (86.12%), Precision (85.91%) and Recall (86.50%).

Table 3 lists the accuracy of vascular structure extraction using different methods. The RPC-Net achieved average skeleton completeness, validity and their combined index F1 Score of 85.50%, 92.38% and 88.70%, respectively, which all outperformed the other methods.

Fig. 4 illustrates the 3D visualization of cerebrovascular segmentation and skeleton extraction by each method. The red, green and blue colors mark the true positives, false positives and false negatives, respectively. False negatives are displayed with an opacity of 0.2 to prevent obscuring true positives. For the skeleton extraction, the gray skeleton represents the ground truth, while the green and blue colors mark false positives and false negatives, respectively. The results indicate that all the methods obtained overall good intracranial vessel

Table 2
Quantitative comparison of different deep learning methods on cerebrovascular segmentation using four-fold cross validation.

Method	DSC (%)	Precision (%)	Recall (%)
3D U-Net	82.19 ± 3.27	82.27 ± 4.97	82.35 ± 3.87
BRAVE-Net	82.12 ± 3.17	80.88 ± 5.13	83.64 ± 3.19
VC-Net	84.19 ± 2.88	84.57 ± 4.08	83.99 ± 3.66
RPC-Net	**86.12 ± 2.75**	**85.91 ± 4.16**	**86.50 ± 3.14**

Table 3
Quantitative comparison of different deep learning methods on vascular structure using four-fold cross validation.

Method	Completeness (%)	Validity (%)	F1 Score (%)
3D U-Net	81.01 ± 6.37	86.53 ± 6.08	83.47 ± 4.82
BRAVE-Net	83.13 ± 5.91	85.31 ± 5.67	84.02 ± 4.43
VC-Net	83.73 ± 6.04	89.69 ± 3.89	86.45 ± 3.96
RPC-Net	**85.50 ± 5.25**	**92.38 ± 3.36**	**88.70 ± 3.48**

segmentation, with most vessels extracted and few over-segmentation errors of scalp vessels. However, the proposed RPC-Net achieved more complete results, where vessel rupture and under-segmentation errors are significantly reduced compared with other methods.

3.5. Performance on vessels of different radii and intensities

The size and intensity of intracranial vessels have large variability, which may affect the performance and surgical planning. To evaluate the accuracy of different methods on vessels of different radii and intensities, skeleton analysis method (Nunez-Iglesias et al., 2018) was adopted to divide the overall vessel into branches according to the skeleton endpoints. Considering each vessel branch as a cylinder, the equivalent radius R_{b_i} of the branch b_i was calculated as

$$R_{b_i} = \sqrt{\frac{V_{b_i}}{\pi L_{b_i}}}, \tag{15}$$

where V_{b_i} and L_{b_i} represent the volume and skeletal length of b_i, respectively. The branch intensity I_{b_i} was defined as the average intensity of the branch voxel:

$$I_{b_i} = \frac{\sum_{v \in b_i} I(v)}{N_{b_i}}, \tag{16}$$

where $I(v)$ denotes the intensity of voxel v and N_{b_i} denotes the voxel quantity of b_i. Furthermore, the branch radii were divided into five groups with an interval of 1 mm, and the Group 1–5 respectively represent 0–1 mm, 1–2 mm, 2–3 mm, 3–4 mm and more than 4 mm. The intensities of all branches of each case were normalized to [0,1] and divided into Group 1–5 at an interval of 0.2. Fig. 5 shows the segmentation DSC of each group of vessel branches. The results show that with the increase in vessel radius and intensity, the segmentation accuracy of each method tended to increase. The four methods achieved good segmentation for larger and higher-intensity vessels. However, RPC-Net outperformed other methods on the thinner and darker vessels,

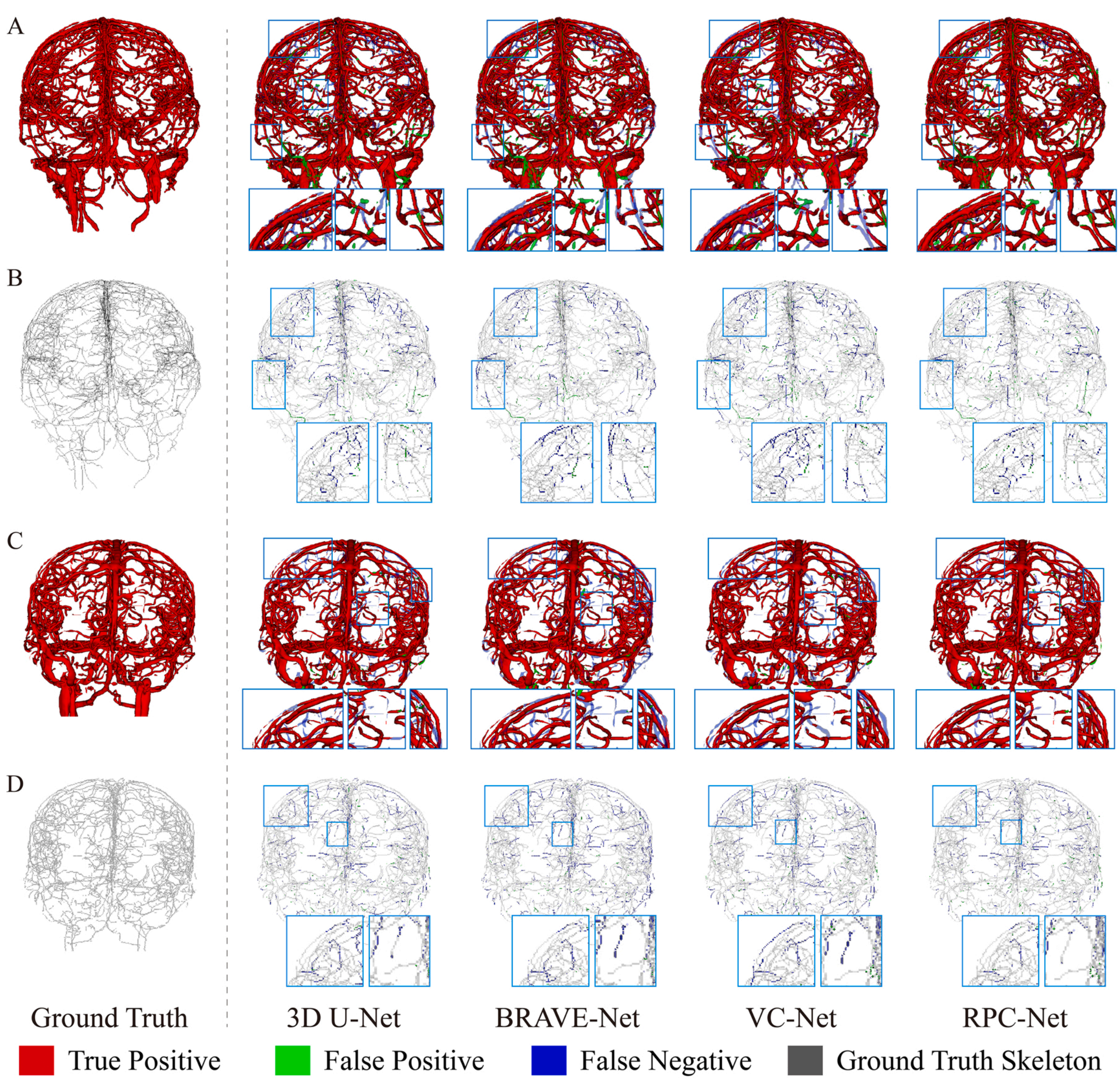

Fig. 4. 3D illustration of cerebrovascular segmentation results (A and C) and skeleton extraction results (B and D) by different methods. The cyan boxes mark the reduced errors compared to other methods in the RPC-Net results.

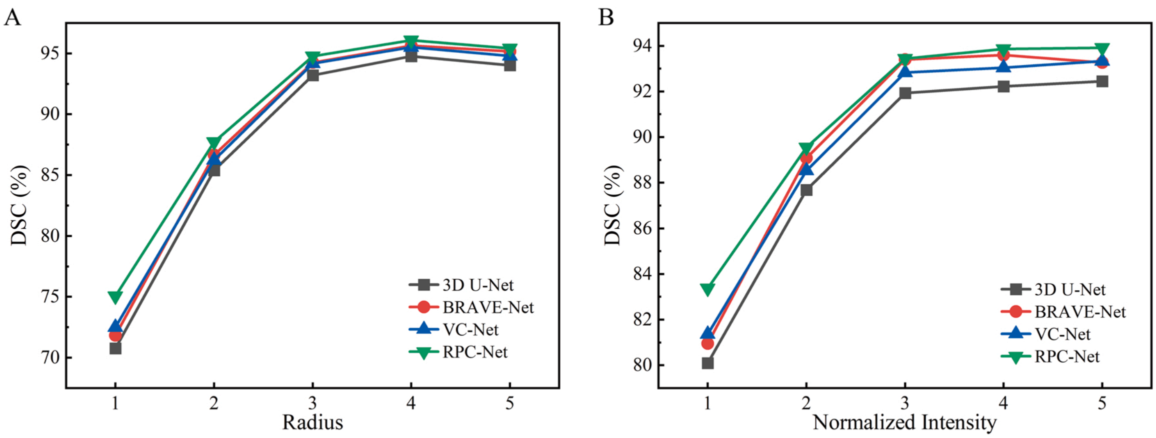

Fig. 5. The Segmentation DSC of different methods on different radius (A) and intensity (B) group of vessel branches.

especially those with the smallest radii and lowest intensities.

Fig. 6 illustrates the visualization of the segmentation on image slices with vessels of different radii and intensities. The results indicate that the methods all achieved desirable results on vessels with large radii and high intensity (Fig. 6(A)). However, the thin and low-intensity intracranial vessels (Fig. 6(B-C)) were confused with the background. They were not dominant compared with the large high-intensity vessels in the image, resulting in poor segmentation. On such vessels, 3D U-Net, BRAVE-Net and VC-Net had more under-segmentation errors, manifested as vessel rupture and missing. In contrast, the segmentation of RPC-Net was more complete, with under-segmentation errors suppressed.

3.6. Impact of patch size and projection direction

Patch size and projection direction will affect the presentation of vessels in the projections, which may affect the segmentation performance. We selected four kinds of patch sizes and six kinds of projection angle quantities and tested the performances of the networks with these parameters (Table 4). The result shows that the small patch size led to a drop in segmentation accuracy, while larger-size input consumed more computing resources. As the number of projection angles increased, the accuracy first increased and then decreased. When the number of projection angles was 96, that is, the projection direction interval was about 2°, the network obtained the highest DSC. Therefore, considering the segmentation performance and computation consumption, the patch size of 64 × 64 × 64 and projection angle quantity of 96 were adopted as the final model parameters.

3.7. Evaluation of applications in surgical planning

To evaluate the application of the method in surgical planning, we selected one case for trajectory planning. Two targets were set in the deep brain, representing tumor ablation sites or electrode sampling

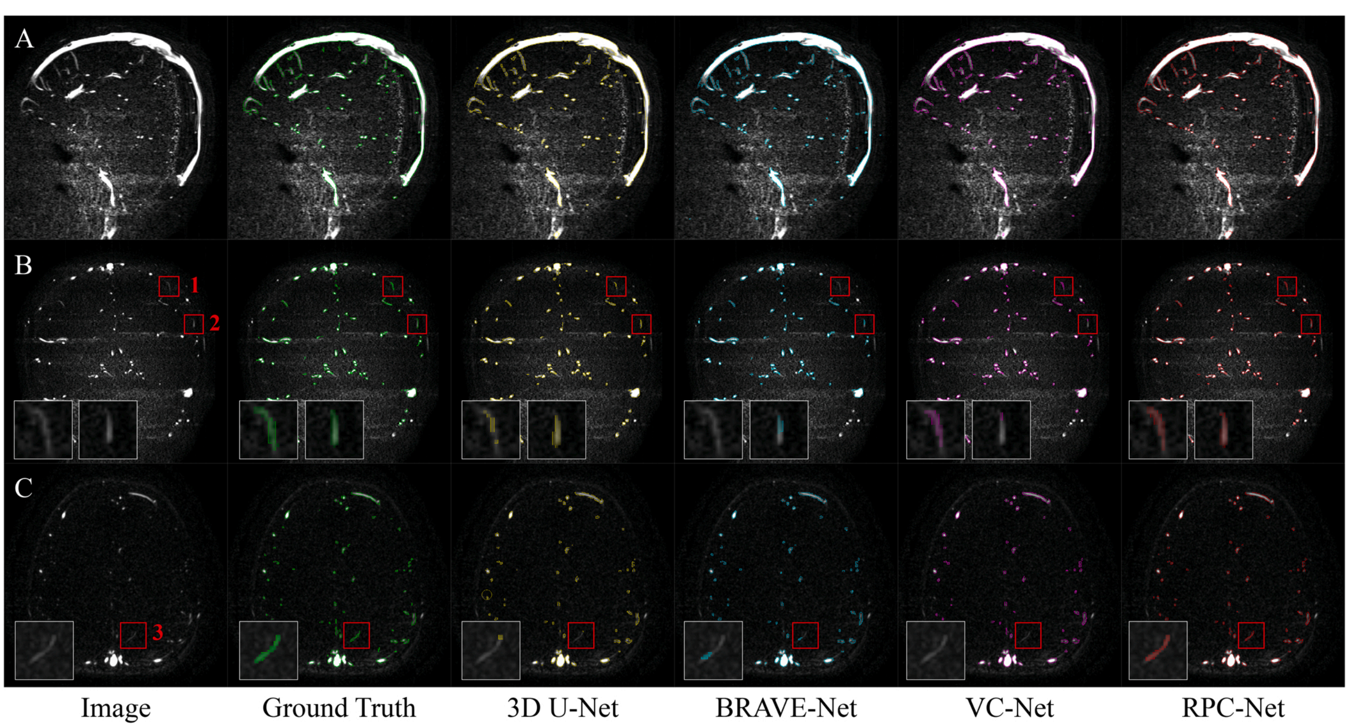

Fig. 6. The visualization of the segmentation on image slices with vessels of different radii and intensities. A. Vessels with large radii and high intensity. B, C. Vessels with smaller radii and lower intensity, where the normalized intensity of vessels 1, 2, and 3 were 0.0359, 0.0700, and 0.0389, respectively, and the radii were 0.66 mm, 0.97 mm, and 1.03 mm, respectively.

Table 4
The performances and computation resource (batch size = 4, test phase) of RPC-Net with different patch sizes and projection angle quantities.

Patch Size	Num. of Projection	Angle Interval (°)	Memory (MB)	DSC (%)
(64, 64, 64)	32	5.63	3920	85.02
(64, 64, 64)	64	2.81	5502	86.06
(64, 64, 64)	**96**	**1.88**	**7178**	**86.12**
(64, 64, 64)	128	1.41	8562	85.85
(64, 64, 64)	160	1.13	10166	85.67
(64, 64, 64)	192	0.94	11660	85.62
(96, 96, 96)	96	1.88	18330	86.17
(48, 48, 48)	96	1.88	5298	84.94
(32, 32, 32)	96	1.88	2998	81.20

areas. Prompted by the ground truth of the vessel structure, four trajectories were manually placed for each target. All trajectories were represented by cylinders with a radius of 0.86 mm, named T_{1-1}, …, T_{1-4}, T_{2-1}, …, T_{2-4} (Fig. 7). T_{1-4} and T_{2-4} intersected with vessels while the others were away from vessels. Table 5 lists the closest distances from these trajectories to the manually annotated and the predicted vessel structure. The result shows that the two closest distances were similar, with errors within 0.4 mm. In addition, when the trajectories (T_{1-4} and T_{2-4}) intersected with vessels, the closest distance to the predicted result was less than 0, indicating that the contacts between the vessels and trajectories were detected.

4. Discussion

In this paper, a cerebrovascular segmentation method for PC-MRA named RPC-Net is proposed to address the challenge of the vascular complex topology and distribution sparsity. The method introduces multi-directional projections to strengthen the sparse 3D vascular representation and highlight the vascular topological connectivity. The mutual transform between the 3D image and the projection domain is constructed through the Radon transform and its inverse transform that

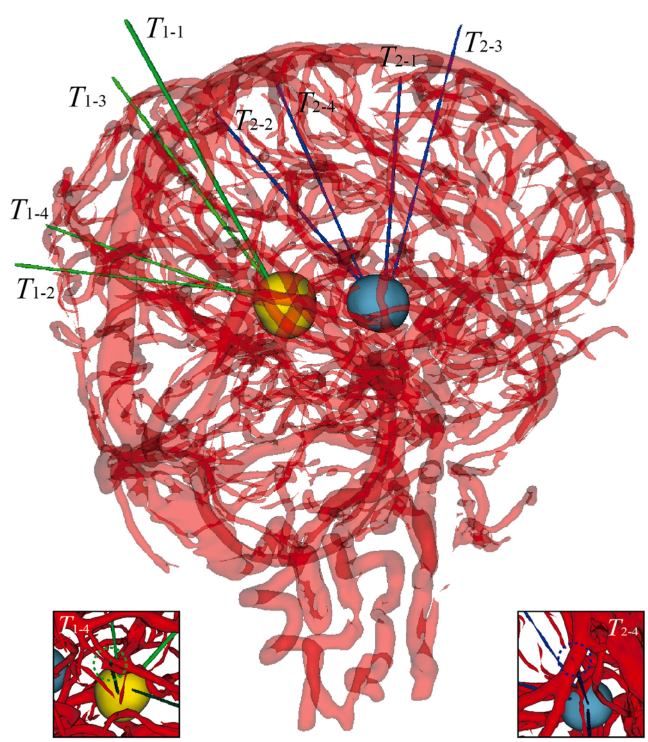

Fig. 7. Application of vascular structure in surgical planning.

Table 5
The comparison of the closest distances from these trajectories to the ground truth and predicted vessel structure.

Trajectory	Minimum Distance to Ground Truth (mm)	Minimum Distance to Prediction (mm)	Error (mm)
T_{1-1}	2.078	2.474	0.396
T_{1-2}	2.474	2.750	0.276
T_{1-3}	2.750	3.000	0.250
$\boldsymbol{T_{1-4}}$	**-0.600**	**-0.600**	**0**
T_{2-1}	2.078	2.245	0.167
T_{2-2}	0.600	0.600	0
T_{2-3}	1.470	1.342	-0.128
$\boldsymbol{T_{2-4}}$	**-1.039**	**-0.849**	**0.190**

commonly used in CT reconstruction. A four-fold cross validation experiment was performed on the local dataset, where RPC-Net obtained the average Dice similarity coefficient, precision and recall of 86.12%, 85.91% and 86.50%, respectively. The average completeness and validity of the vascular structure achieved 85.50% and 92.38%, respectively.

This paper compared the RPC-Net with 3D U-Net, BRAVE-Net and VC-Net. The reference methods were regarded as the baseline model, the representative of the U-Net variant model and the model that also uses the projection, respectively. The experiment result shows that RPC-Net was superior to other methods in cerebrovascular segmentation accuracy and the completeness and validity of the vessel skeleton, with vessel rupture errors and under-segmentation significantly reduced. This may be because that the topological information implied by the multi-directional projections enhances the ability of the network to identify vessels with complex topology. In contrast, 3D U-Net and BRAVE-Net lack vascular topological hints. VC-Net enhances vascular spatial distribution probability and improves the image signal-to-noise ratio through MIP. However, the single-directional projection loses extensive information and may lead to the problem of the overlap between different vessels, resulting in a decrease in accuracy.

The radius and intensity of the vessels have significant variability. Small and low-intensity vessels are generally prone to under-segmentation. Such vessels are mainly distributed in the brain sulci and deep brain, which are also key risk areas for neurosurgery. The experimental results in this paper show that the segmentation accuracy on small and low-intensity vessels was significantly lower than that on large and high-intensity vessels. The network mainly focused on large and bright vessels, as vessels with more voxels and higher intensity had prominent features. Our work improved on small and dark vessels, with the under-segmentation errors suppressed. The possible reason is that the introduced multi-directional projection suggests the topological connection between dark, small vessels and their main branches, making the network more biased towards analyzing topological features. The projected image size and the projection direction determine the characteristics of vessels in the projection images, which may affect the segmentation accuracy. As more directions of the projections were adopted, the accuracy of cerebrovascular segmentation first increased and then decreased. When the projected image and the field of view of the network became smaller, the segmentation performance degraded. In terms of projection angle, the line integral operation in the Radon transform may cause the vessel intensity to be averaged and darkened by the background voxels, while the projections of more directions may enhance the probability of these vessels appearing in the projection images. In addition, the shape and topological connectivity of all vessels are more fully characterized by more projections, thus improving the segmentation accuracy. However, redundant projection input may cause the network to be unable to effectively extract information, resulting in a decrease in accuracy. In terms of input size, large-field-of-view images can reflect a more comprehensive topology, which is beneficial to vessel segmentation.

This work has potential application in assisting neurosurgery planning. The vessel structure extracted by the proposed method has high accuracy and completeness. The trajectory planning experiment also demonstrates that the constructed vessel model can accurately calculate the distance between the placed trajectories and the vessels while detecting the contacts between them. Therefore, this work can provide a reliable intracranial vascular model for minimally invasive stereotactic neurosurgery requiring device implantation into the brain. Moreover, since the proposed method obtains more accurate vessel segmentation with fewer under-segmentation errors compared to other methods, it has the potential advantage of reducing the risk of planned paths intersecting vessels.

There are still some limitations in this work. The intensity averaging effect of the Radon transform limits the performance of the network on small dark vessels and some under-segmentation errors still occurred. To this end, we will focus on enhancing the vessels in both the original and the projection images to improve the signal of small low-intensity vessels. The problem of local fractures caused by low-velocity vessels still needs to be solved. Although Chen et al. have tried to address this problem based on the smoothness of vessel curvature (Chen et al., 2022), the method is not suitable for vessels with significant changes in curvature. In the future, we will conduct further research on this issue to obtain a more complete vascular network. Moreover, other back-projection algorisms, such as iterative optimization methods, will be tried and assessed in the future work. The dataset used in this paper is still insufficient in quantity and proportion, so more data from different centers will be introduced to verify the performance of the method.

5. Conclusions

This paper proposes an RPC-Net for cerebrovascular segmentation in PC-MRA to address the problem of vascular complex topology and sparse distribution. The RPC-Net introduces multi-directional Radon projections of the original images and uses a two-stream network to learn the features of the 3D image and the projection, aiming to strengthen the sparse 3D vascular representation and highlight the vascular topological connectivity. The mapping between the image domain and the projection domain is constructed by the Radon transform and its back-projection transform. Finally, the image-projection joint features are obtained and used to predict vessel voxels. The method was assessed on a local dataset containing 128 PC-MRA scans. The results demonstrate that RPC-Net achieves an accurate and complete segmentation of intracranial vessels and is superior to existing methods, with the potential to assist in neurosurgery planning.

Ethical approval

For this type of study formal consent is not required. The testing data were collected at our institution with approval from the institutional review board.

CRediT authorship contribution statement

Wenhai Weng: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing. **Hui Ding:** Conceptualization, Supervision, Project administration, Funding acquisition. **Jianjun Bai:** Investigation, Resources, Data curation. **Wenjing Zhou:** Investigation, Resources, Data curation. **Guangzhi Wang:** Conceptualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Guangzhi Wang reports equipment, drugs, or supplies was provided by The Second Hospital of Hebei Medical Uniyersity. Guangzhi Wang reports equipment, drugs, or supplies was provided by Peking University First Hospital. Guangzhi Wang reports equipment, drugs, or supplies was provided by Beijing RIMAG Imaging Center.

Data availability

The authors do not have permission to share data.

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